IBScovars_application_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
Complete data	0.0	Complete−data benchmark	1 vs 0	0.285	0.125	−0.023	0.592	0.615
Complete data	0.0	Complete−data benchmark	1 vs 0	0.285	0.125	−0.022	0.591	0.612
Complete data	0.0	Complete−data benchmark	2 vs 0	0.297	0.115	0.015	0.579	0.564
Complete data	0.0	Complete−data benchmark	2 vs 0	0.297	0.126	−0.011	0.605	0.617
Complete data	0.0	Complete−data benchmark	3 vs 0	0.351	0.123	0.049	0.653	0.604
Complete data	0.0	Complete−data benchmark	3 vs 0	0.351	0.127	0.039	0.663	0.624
Complete data	0.0	Complete−data benchmark	4 vs 0	0.348	0.126	0.038	0.658	0.620
Complete data	0.0	Complete−data benchmark	4 vs 0	0.348	0.127	0.037	0.659	0.622
MCAR 10% (achieved 10%)	0.1	Complete−case	1 vs 0	0.335	0.133	0.007	0.664	0.658
MCAR 10% (achieved 10%)	0.1	Complete−case	1 vs 0	0.335	0.133	0.009	0.661	0.652
MCAR 10% (achieved 10%)	0.1	Complete−case	2 vs 0	0.326	0.120	0.030	0.622	0.593
MCAR 10% (achieved 10%)	0.1	Complete−case	2 vs 0	0.326	0.132	0.002	0.650	0.648
MCAR 10% (achieved 10%)	0.1	Complete−case	3 vs 0	0.370	0.130	0.048	0.692	0.643
MCAR 10% (achieved 10%)	0.1	Complete−case	3 vs 0	0.370	0.134	0.040	0.700	0.660
MCAR 10% (achieved 10%)	0.1	Complete−case	4 vs 0	0.445	0.130	0.124	0.766	0.642
MCAR 10% (achieved 10%)	0.1	Complete−case	4 vs 0	0.445	0.134	0.117	0.774	0.657
MCAR 10% (achieved 10%)	0.1	PMM MI	1 vs 0	0.335	0.132	0.010	0.659	0.649
MCAR 10% (achieved 10%)	0.1	PMM MI	1 vs 0	0.335	0.131	0.013	0.656	0.644
MCAR 10% (achieved 10%)	0.1	PMM MI	2 vs 0	0.325	0.117	0.038	0.613	0.575
MCAR 10% (achieved 10%)	0.1	PMM MI	2 vs 0	0.325	0.126	0.017	0.633	0.616

IBScovars_application_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
MCAR 10% (achieved 10%)	0.1	PMM MI	3 vs 0	0.365	0.128	0.052	0.679	0.627
MCAR 10% (achieved 10%)	0.1	PMM MI	3 vs 0	0.365	0.131	0.045	0.686	0.641
MCAR 10% (achieved 10%)	0.1	PMM MI	4 vs 0	0.448	0.128	0.133	0.763	0.629
MCAR 10% (achieved 10%)	0.1	PMM MI	4 vs 0	0.448	0.131	0.128	0.768	0.641
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	1 vs 0	0.331	0.134	0.003	0.659	0.657
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	1 vs 0	0.331	0.133	0.006	0.656	0.651
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	2 vs 0	0.336	0.119	0.043	0.628	0.585
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	2 vs 0	0.336	0.128	0.022	0.649	0.627
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	3 vs 0	0.376	0.129	0.059	0.693	0.633
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	3 vs 0	0.376	0.132	0.053	0.699	0.647
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	4 vs 0	0.451	0.130	0.132	0.770	0.638
MCAR 10% (achieved 10%)	0.1	Propensity−score MI	4 vs 0	0.451	0.133	0.126	0.775	0.649
MCAR 10% (achieved 10%)	0.1	Regression MI	1 vs 0	0.336	0.137	0.001	0.671	0.670
MCAR 10% (achieved 10%)	0.1	Regression MI	1 vs 0	0.336	0.136	0.003	0.668	0.666
MCAR 10% (achieved 10%)	0.1	Regression MI	2 vs 0	0.320	0.121	0.023	0.618	0.595
MCAR 10% (achieved 10%)	0.1	Regression MI	2 vs 0	0.320	0.130	0.002	0.639	0.637
MCAR 10% (achieved 10%)	0.1	Regression MI	3 vs 0	0.374	0.134	0.045	0.703	0.657
MCAR 10% (achieved 10%)	0.1	Regression MI	3 vs 0	0.374	0.137	0.039	0.709	0.669
MCAR 10% (achieved 10%)	0.1	Regression MI	4 vs 0	0.446	0.136	0.112	0.780	0.667
MCAR 10% (achieved 10%)	0.1	Regression MI	4 vs 0	0.446	0.139	0.107	0.785	0.678

IBScovars_application_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
MCAR 20% (achieved 20%)	0.2	Complete−case	1 vs 0	0.250	0.143	−0.103	0.603	0.706
MCAR 20% (achieved 20%)	0.2	Complete−case	1 vs 0	0.250	0.144	−0.103	0.604	0.707
MCAR 20% (achieved 20%)	0.2	Complete−case	2 vs 0	0.281	0.132	−0.045	0.607	0.652
MCAR 20% (achieved 20%)	0.2	Complete−case	2 vs 0	0.281	0.145	−0.075	0.638	0.713
MCAR 20% (achieved 20%)	0.2	Complete−case	3 vs 0	0.314	0.143	−0.039	0.667	0.707
MCAR 20% (achieved 20%)	0.2	Complete−case	3 vs 0	0.314	0.146	−0.045	0.673	0.719
MCAR 20% (achieved 20%)	0.2	Complete−case	4 vs 0	0.332	0.143	−0.020	0.684	0.705
MCAR 20% (achieved 20%)	0.2	Complete−case	4 vs 0	0.332	0.146	−0.027	0.691	0.719
MCAR 20% (achieved 20%)	0.2	PMM MI	1 vs 0	0.241	0.141	−0.104	0.586	0.690
MCAR 20% (achieved 20%)	0.2	PMM MI	1 vs 0	0.241	0.142	−0.105	0.587	0.692
MCAR 20% (achieved 20%)	0.2	PMM MI	2 vs 0	0.261	0.127	−0.051	0.572	0.622
MCAR 20% (achieved 20%)	0.2	PMM MI	2 vs 0	0.261	0.136	−0.072	0.593	0.665
MCAR 20% (achieved 20%)	0.2	PMM MI	3 vs 0	0.298	0.142	−0.049	0.646	0.695
MCAR 20% (achieved 20%)	0.2	PMM MI	3 vs 0	0.298	0.144	−0.054	0.651	0.704
MCAR 20% (achieved 20%)	0.2	PMM MI	4 vs 0	0.311	0.138	−0.028	0.650	0.678
MCAR 20% (achieved 20%)	0.2	PMM MI	4 vs 0	0.311	0.141	−0.032	0.655	0.688
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	1 vs 0	0.250	0.140	−0.094	0.594	0.687
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	1 vs 0	0.250	0.141	−0.093	0.593	0.686
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	2 vs 0	0.289	0.136	−0.044	0.621	0.664
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	2 vs 0	0.289	0.144	−0.062	0.639	0.702

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scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	3 vs 0	0.302	0.145	−0.054	0.658	0.712
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	3 vs 0	0.302	0.147	−0.057	0.662	0.719
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	4 vs 0	0.336	0.140	−0.008	0.679	0.687
MCAR 20% (achieved 20%)	0.2	Propensity−score MI	4 vs 0	0.336	0.143	−0.012	0.684	0.696
MCAR 20% (achieved 20%)	0.2	Regression MI	1 vs 0	0.255	0.146	−0.103	0.613	0.715
MCAR 20% (achieved 20%)	0.2	Regression MI	1 vs 0	0.255	0.146	−0.103	0.613	0.716
MCAR 20% (achieved 20%)	0.2	Regression MI	2 vs 0	0.275	0.134	−0.053	0.604	0.657
MCAR 20% (achieved 20%)	0.2	Regression MI	2 vs 0	0.275	0.142	−0.074	0.624	0.697
MCAR 20% (achieved 20%)	0.2	Regression MI	3 vs 0	0.316	0.143	−0.036	0.668	0.705
MCAR 20% (achieved 20%)	0.2	Regression MI	3 vs 0	0.316	0.146	−0.040	0.673	0.713
MCAR 20% (achieved 20%)	0.2	Regression MI	4 vs 0	0.326	0.149	−0.040	0.692	0.733
MCAR 20% (achieved 20%)	0.2	Regression MI	4 vs 0	0.326	0.151	−0.043	0.695	0.738
MCAR 30% (achieved 30%)	0.3	Complete−case	1 vs 0	0.407	0.153	0.031	0.783	0.752
MCAR 30% (achieved 30%)	0.3	Complete−case	1 vs 0	0.407	0.146	0.048	0.765	0.717
MCAR 30% (achieved 30%)	0.3	Complete−case	2 vs 0	0.364	0.135	0.033	0.695	0.662
MCAR 30% (achieved 30%)	0.3	Complete−case	2 vs 0	0.364	0.143	0.013	0.715	0.702
MCAR 30% (achieved 30%)	0.3	Complete−case	3 vs 0	0.273	0.139	−0.068	0.614	0.682
MCAR 30% (achieved 30%)	0.3	Complete−case	3 vs 0	0.273	0.146	−0.084	0.630	0.715
MCAR 30% (achieved 30%)	0.3	Complete−case	4 vs 0	0.393	0.148	0.029	0.756	0.727
MCAR 30% (achieved 30%)	0.3	Complete−case	4 vs 0	0.393	0.147	0.032	0.754	0.721

IBScovars_application_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
MCAR 30% (achieved 30%)	0.3	PMM MI	1 vs 0	0.443	0.147	0.083	0.804	0.721
MCAR 30% (achieved 30%)	0.3	PMM MI	1 vs 0	0.443	0.144	0.090	0.797	0.707
MCAR 30% (achieved 30%)	0.3	PMM MI	2 vs 0	0.381	0.128	0.068	0.694	0.626
MCAR 30% (achieved 30%)	0.3	PMM MI	2 vs 0	0.381	0.132	0.057	0.705	0.648
MCAR 30% (achieved 30%)	0.3	PMM MI	3 vs 0	0.275	0.128	−0.039	0.590	0.629
MCAR 30% (achieved 30%)	0.3	PMM MI	3 vs 0	0.275	0.131	−0.046	0.597	0.643
MCAR 30% (achieved 30%)	0.3	PMM MI	4 vs 0	0.419	0.142	0.070	0.768	0.698
MCAR 30% (achieved 30%)	0.3	PMM MI	4 vs 0	0.419	0.141	0.072	0.766	0.694
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	1 vs 0	0.419	0.158	0.032	0.806	0.773
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	1 vs 0	0.419	0.156	0.039	0.799	0.761
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	2 vs 0	0.386	0.139	0.047	0.726	0.679
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	2 vs 0	0.386	0.143	0.038	0.735	0.697
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	3 vs 0	0.261	0.136	−0.072	0.593	0.666
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	3 vs 0	0.261	0.139	−0.079	0.600	0.680
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	4 vs 0	0.403	0.161	0.009	0.797	0.788
MCAR 30% (achieved 30%)	0.3	Propensity−score MI	4 vs 0	0.403	0.161	0.008	0.797	0.789
MCAR 30% (achieved 30%)	0.3	Regression MI	1 vs 0	0.459	0.153	0.085	0.833	0.748
MCAR 30% (achieved 30%)	0.3	Regression MI	1 vs 0	0.459	0.150	0.093	0.825	0.733
MCAR 30% (achieved 30%)	0.3	Regression MI	2 vs 0	0.402	0.146	0.045	0.759	0.714
MCAR 30% (achieved 30%)	0.3	Regression MI	2 vs 0	0.402	0.149	0.037	0.767	0.730

IBScovars_application_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
MCAR 30% (achieved 30%)	0.3	Regression MI	3 vs 0	0.292	0.141	−0.052	0.637	0.689
MCAR 30% (achieved 30%)	0.3	Regression MI	3 vs 0	0.292	0.143	−0.057	0.642	0.700
MCAR 30% (achieved 30%)	0.3	Regression MI	4 vs 0	0.436	0.164	0.034	0.839	0.804
MCAR 30% (achieved 30%)	0.3	Regression MI	4 vs 0	0.436	0.163	0.037	0.836	0.800
MCAR 40% (achieved 40%)	0.4	Complete−case	1 vs 0	0.340	0.166	−0.070	0.750	0.819
MCAR 40% (achieved 40%)	0.4	Complete−case	1 vs 0	0.340	0.164	−0.063	0.743	0.806
MCAR 40% (achieved 40%)	0.4	Complete−case	2 vs 0	0.211	0.160	−0.183	0.604	0.787
MCAR 40% (achieved 40%)	0.4	Complete−case	2 vs 0	0.211	0.166	−0.196	0.618	0.814
MCAR 40% (achieved 40%)	0.4	Complete−case	3 vs 0	0.279	0.162	−0.122	0.679	0.801
MCAR 40% (achieved 40%)	0.4	Complete−case	3 vs 0	0.279	0.167	−0.133	0.691	0.823
MCAR 40% (achieved 40%)	0.4	Complete−case	4 vs 0	0.374	0.159	−0.017	0.765	0.782
MCAR 40% (achieved 40%)	0.4	Complete−case	4 vs 0	0.374	0.167	−0.036	0.783	0.819
MCAR 40% (achieved 40%)	0.4	PMM MI	1 vs 0	0.372	0.142	0.022	0.721	0.699
MCAR 40% (achieved 40%)	0.4	PMM MI	1 vs 0	0.372	0.140	0.028	0.716	0.688
MCAR 40% (achieved 40%)	0.4	PMM MI	2 vs 0	0.232	0.146	−0.127	0.591	0.718
MCAR 40% (achieved 40%)	0.4	PMM MI	2 vs 0	0.232	0.148	−0.133	0.596	0.729
MCAR 40% (achieved 40%)	0.4	PMM MI	3 vs 0	0.289	0.142	−0.060	0.638	0.698
MCAR 40% (achieved 40%)	0.4	PMM MI	3 vs 0	0.289	0.145	−0.067	0.646	0.713
MCAR 40% (achieved 40%)	0.4	PMM MI	4 vs 0	0.399	0.140	0.054	0.744	0.690
MCAR 40% (achieved 40%)	0.4	PMM MI	4 vs 0	0.399	0.144	0.045	0.753	0.708

IBScovars_application_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	1 vs 0	0.325	0.169	−0.089	0.740	0.830
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	1 vs 0	0.325	0.168	−0.086	0.737	0.823
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	2 vs 0	0.212	0.160	−0.181	0.605	0.786
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	2 vs 0	0.212	0.162	−0.185	0.609	0.795
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	3 vs 0	0.286	0.159	−0.104	0.677	0.780
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	3 vs 0	0.286	0.161	−0.108	0.681	0.790
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	4 vs 0	0.361	0.149	−0.006	0.729	0.735
MCAR 40% (achieved 40%)	0.4	Propensity−score MI	4 vs 0	0.361	0.153	−0.016	0.738	0.754
MCAR 40% (achieved 40%)	0.4	Regression MI	1 vs 0	0.379	0.165	−0.028	0.786	0.814
MCAR 40% (achieved 40%)	0.4	Regression MI	1 vs 0	0.379	0.164	−0.025	0.783	0.809
MCAR 40% (achieved 40%)	0.4	Regression MI	2 vs 0	0.236	0.159	−0.155	0.628	0.783
MCAR 40% (achieved 40%)	0.4	Regression MI	2 vs 0	0.236	0.161	−0.161	0.633	0.794
MCAR 40% (achieved 40%)	0.4	Regression MI	3 vs 0	0.301	0.146	−0.058	0.660	0.718
MCAR 40% (achieved 40%)	0.4	Regression MI	3 vs 0	0.301	0.148	−0.065	0.666	0.731
MCAR 40% (achieved 40%)	0.4	Regression MI	4 vs 0	0.376	0.153	−0.003	0.754	0.757
MCAR 40% (achieved 40%)	0.4	Regression MI	4 vs 0	0.376	0.157	−0.011	0.763	0.774

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
FALSE	NA	Non−shrinkage	0.285	0.000	0.000	TRUE
FALSE	0.974	Shrinkage	0.285	0.000	0.000	TRUE
TRUE	NA	Non−shrinkage	0.297	0.000	0.000	TRUE
FALSE	0.974	Shrinkage	0.297	0.000	0.000	TRUE
TRUE	NA	Non−shrinkage	0.351	0.000	0.000	TRUE
TRUE	0.974	Shrinkage	0.351	0.000	0.000	TRUE
TRUE	NA	Non−shrinkage	0.348	0.000	0.000	TRUE
TRUE	0.974	Shrinkage	0.348	0.000	0.000	TRUE
TRUE	NA	Non−shrinkage	0.285	0.051	0.051	TRUE
TRUE	0.846	Shrinkage	0.285	0.051	0.051	TRUE
TRUE	NA	Non−shrinkage	0.297	0.029	0.029	TRUE
TRUE	0.846	Shrinkage	0.297	0.029	0.029	TRUE
TRUE	NA	Non−shrinkage	0.351	0.019	0.019	TRUE
TRUE	0.846	Shrinkage	0.351	0.019	0.019	TRUE
TRUE	NA	Non−shrinkage	0.348	0.097	0.097	TRUE
TRUE	0.846	Shrinkage	0.348	0.097	0.097	TRUE
TRUE	NA	Non−shrinkage	0.285	0.050	0.050	TRUE
TRUE	0.664	Shrinkage	0.285	0.050	0.050	TRUE
TRUE	NA	Non−shrinkage	0.297	0.028	0.028	TRUE
TRUE	0.664	Shrinkage	0.297	0.028	0.028	TRUE

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
TRUE	NA	Non−shrinkage	0.351	0.015	0.015	TRUE
TRUE	0.664	Shrinkage	0.351	0.015	0.015	TRUE
TRUE	NA	Non−shrinkage	0.348	0.100	0.100	TRUE
TRUE	0.664	Shrinkage	0.348	0.100	0.100	TRUE
TRUE	NA	Non−shrinkage	0.285	0.046	0.046	TRUE
TRUE	0.733	Shrinkage	0.285	0.046	0.046	TRUE
TRUE	NA	Non−shrinkage	0.297	0.039	0.039	TRUE
TRUE	0.733	Shrinkage	0.297	0.039	0.039	TRUE
TRUE	NA	Non−shrinkage	0.351	0.025	0.025	TRUE
TRUE	0.733	Shrinkage	0.351	0.025	0.025	TRUE
TRUE	NA	Non−shrinkage	0.348	0.103	0.103	TRUE
TRUE	0.733	Shrinkage	0.348	0.103	0.103	TRUE
TRUE	NA	Non−shrinkage	0.285	0.051	0.051	TRUE
TRUE	0.717	Shrinkage	0.285	0.051	0.051	TRUE
TRUE	NA	Non−shrinkage	0.297	0.024	0.024	TRUE
TRUE	0.717	Shrinkage	0.297	0.024	0.024	TRUE
TRUE	NA	Non−shrinkage	0.351	0.023	0.023	TRUE
TRUE	0.717	Shrinkage	0.351	0.023	0.023	TRUE
TRUE	NA	Non−shrinkage	0.348	0.098	0.098	TRUE
TRUE	0.717	Shrinkage	0.348	0.098	0.098	TRUE

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
FALSE	NA	Non−shrinkage	0.285	−0.034	0.034	TRUE
FALSE	1.000	Shrinkage	0.285	−0.034	0.034	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.016	0.016	TRUE
FALSE	1.000	Shrinkage	0.297	−0.016	0.016	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.037	0.037	TRUE
FALSE	1.000	Shrinkage	0.351	−0.037	0.037	TRUE
FALSE	NA	Non−shrinkage	0.348	−0.016	0.016	TRUE
FALSE	1.000	Shrinkage	0.348	−0.016	0.016	TRUE
FALSE	NA	Non−shrinkage	0.285	−0.043	0.043	TRUE
FALSE	0.835	Shrinkage	0.285	−0.043	0.043	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.036	0.036	TRUE
FALSE	0.835	Shrinkage	0.297	−0.036	0.036	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.052	0.052	TRUE
FALSE	0.835	Shrinkage	0.351	−0.052	0.052	TRUE
FALSE	NA	Non−shrinkage	0.348	−0.036	0.036	TRUE
FALSE	0.835	Shrinkage	0.348	−0.036	0.036	TRUE
FALSE	NA	Non−shrinkage	0.285	−0.035	0.035	TRUE
FALSE	0.846	Shrinkage	0.285	−0.035	0.035	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.008	0.008	TRUE
FALSE	0.846	Shrinkage	0.297	−0.008	0.008	TRUE

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
FALSE	NA	Non−shrinkage	0.351	−0.049	0.049	TRUE
FALSE	0.846	Shrinkage	0.351	−0.049	0.049	TRUE
FALSE	NA	Non−shrinkage	0.348	−0.012	0.012	TRUE
FALSE	0.846	Shrinkage	0.348	−0.012	0.012	TRUE
FALSE	NA	Non−shrinkage	0.285	−0.030	0.030	TRUE
FALSE	0.832	Shrinkage	0.285	−0.030	0.030	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.022	0.022	TRUE
FALSE	0.832	Shrinkage	0.297	−0.022	0.022	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.035	0.035	TRUE
FALSE	0.832	Shrinkage	0.351	−0.035	0.035	TRUE
FALSE	NA	Non−shrinkage	0.348	−0.022	0.022	TRUE
FALSE	0.832	Shrinkage	0.348	−0.022	0.022	TRUE
TRUE	NA	Non−shrinkage	0.285	0.122	0.122	TRUE
TRUE	0.781	Shrinkage	0.285	0.122	0.122	TRUE
TRUE	NA	Non−shrinkage	0.297	0.067	0.067	TRUE
TRUE	0.781	Shrinkage	0.297	0.067	0.067	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.078	0.078	TRUE
FALSE	0.781	Shrinkage	0.351	−0.078	0.078	TRUE
TRUE	NA	Non−shrinkage	0.348	0.045	0.045	TRUE
TRUE	0.781	Shrinkage	0.348	0.045	0.045	TRUE

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
TRUE	NA	Non−shrinkage	0.285	0.159	0.159	TRUE
TRUE	0.521	Shrinkage	0.285	0.159	0.159	TRUE
TRUE	NA	Non−shrinkage	0.297	0.084	0.084	TRUE
TRUE	0.521	Shrinkage	0.297	0.084	0.084	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.075	0.075	TRUE
FALSE	0.521	Shrinkage	0.351	−0.075	0.075	TRUE
TRUE	NA	Non−shrinkage	0.348	0.071	0.071	TRUE
TRUE	0.521	Shrinkage	0.348	0.071	0.071	TRUE
TRUE	NA	Non−shrinkage	0.285	0.134	0.134	TRUE
TRUE	0.484	Shrinkage	0.285	0.134	0.134	TRUE
TRUE	NA	Non−shrinkage	0.297	0.090	0.090	TRUE
TRUE	0.484	Shrinkage	0.297	0.090	0.090	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.090	0.090	TRUE
FALSE	0.484	Shrinkage	0.351	−0.090	0.090	TRUE
TRUE	NA	Non−shrinkage	0.348	0.055	0.055	TRUE
TRUE	0.484	Shrinkage	0.348	0.055	0.055	TRUE
TRUE	NA	Non−shrinkage	0.285	0.174	0.174	TRUE
TRUE	0.523	Shrinkage	0.285	0.174	0.174	TRUE
TRUE	NA	Non−shrinkage	0.297	0.105	0.105	TRUE
TRUE	0.523	Shrinkage	0.297	0.105	0.105	TRUE

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
FALSE	NA	Non−shrinkage	0.351	−0.058	0.058	TRUE
FALSE	0.523	Shrinkage	0.351	−0.058	0.058	TRUE
TRUE	NA	Non−shrinkage	0.348	0.088	0.088	TRUE
TRUE	0.523	Shrinkage	0.348	0.088	0.088	TRUE
FALSE	NA	Non−shrinkage	0.285	0.055	0.055	TRUE
FALSE	1.000	Shrinkage	0.285	0.055	0.055	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.086	0.086	TRUE
FALSE	1.000	Shrinkage	0.297	−0.086	0.086	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.072	0.072	TRUE
FALSE	1.000	Shrinkage	0.351	−0.072	0.072	TRUE
FALSE	NA	Non−shrinkage	0.348	0.026	0.026	TRUE
FALSE	1.000	Shrinkage	0.348	0.026	0.026	TRUE
TRUE	NA	Non−shrinkage	0.285	0.087	0.087	TRUE
TRUE	0.874	Shrinkage	0.285	0.087	0.087	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.065	0.065	TRUE
FALSE	0.874	Shrinkage	0.297	−0.065	0.065	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.062	0.062	TRUE
FALSE	0.874	Shrinkage	0.351	−0.062	0.062	TRUE
TRUE	NA	Non−shrinkage	0.348	0.051	0.051	TRUE
TRUE	0.874	Shrinkage	0.348	0.051	0.051	TRUE

IBScovars_application_results.csv

significant	shrink_alpha	variance_method	benchmark	error	abs_error	benchmark_covered
FALSE	NA	Non−shrinkage	0.285	0.041	0.041	TRUE
FALSE	0.846	Shrinkage	0.285	0.041	0.041	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.085	0.085	TRUE
FALSE	0.846	Shrinkage	0.297	−0.085	0.085	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.064	0.064	TRUE
FALSE	0.846	Shrinkage	0.351	−0.064	0.064	TRUE
FALSE	NA	Non−shrinkage	0.348	0.013	0.013	TRUE
FALSE	0.846	Shrinkage	0.348	0.013	0.013	TRUE
FALSE	NA	Non−shrinkage	0.285	0.094	0.094	TRUE
FALSE	0.870	Shrinkage	0.285	0.094	0.094	TRUE
FALSE	NA	Non−shrinkage	0.297	−0.061	0.061	TRUE
FALSE	0.870	Shrinkage	0.297	−0.061	0.061	TRUE
FALSE	NA	Non−shrinkage	0.351	−0.050	0.050	TRUE
FALSE	0.870	Shrinkage	0.351	−0.050	0.050	TRUE
FALSE	NA	Non−shrinkage	0.348	0.028	0.028	TRUE
FALSE	0.870	Shrinkage	0.348	0.028	0.028	TRUE

IBScovars_ci_width_trend.csv

missing_rate	method	variance_method	avg_width	mean_shrink_alpha
0.1	PMM MI	Non−shrinkage	0.620	NA
0.1	PMM MI	Shrinkage	0.635	0.664
0.1	Propensity−score MI	Non−shrinkage	0.628	NA
0.1	Propensity−score MI	Shrinkage	0.643	0.733
0.1	Regression MI	Non−shrinkage	0.647	NA
0.1	Regression MI	Shrinkage	0.662	0.717
0.2	PMM MI	Non−shrinkage	0.671	NA
0.2	PMM MI	Shrinkage	0.687	0.835
0.2	Propensity−score MI	Non−shrinkage	0.688	NA
0.2	Propensity−score MI	Shrinkage	0.701	0.846
0.2	Regression MI	Non−shrinkage	0.702	NA
0.2	Regression MI	Shrinkage	0.716	0.832
0.3	PMM MI	Non−shrinkage	0.669	NA
0.3	PMM MI	Shrinkage	0.673	0.521
0.3	Propensity−score MI	Non−shrinkage	0.727	NA
0.3	Propensity−score MI	Shrinkage	0.731	0.484
0.3	Regression MI	Non−shrinkage	0.739	NA
0.3	Regression MI	Shrinkage	0.740	0.523
0.4	PMM MI	Non−shrinkage	0.701	NA
0.4	PMM MI	Shrinkage	0.710	0.874

IBScovars_ci_width_trend.csv

missing_rate	method	variance_method	avg_width	mean_shrink_alpha
0.4	Propensity−score MI	Non−shrinkage	0.783	NA
0.4	Propensity−score MI	Shrinkage	0.790	0.846
0.4	Regression MI	Non−shrinkage	0.768	NA
0.4	Regression MI	Shrinkage	0.777	0.870

IBScovars_complete_benchmark_contrasts.csv

contrast	complete_data_contrast
1 vs 0	0.285
2 vs 0	0.297
3 vs 0	0.351
4 vs 0	0.348

IBScovars_complete_data_summary.csv

group	n	mean	sd	variance
0	71	0.217	0.695	0.483
1	78	0.502	0.830	0.689
2	75	0.514	0.690	0.476
3	72	0.568	0.771	0.595
4	73	0.565	0.812	0.660

IBScovars_dataset_info.csv

dataset	n_total	n_groups	natural_missing_pct	complete_data_variance_ratio
IBScovars	369	5	0	1.448

IBScovars_primary_30pct_table.csv

scenario	method	variance_method	contrast	estimate	lower	upper	width	significant
MCAR 30% (achieved 30%)	Complete−case	Non−shrinkage	1 vs 0	0.407	0.031	0.783	0.752	TRUE
MCAR 30% (achieved 30%)	Complete−case	Shrinkage	1 vs 0	0.407	0.048	0.765	0.717	TRUE
MCAR 30% (achieved 30%)	Complete−case	Non−shrinkage	2 vs 0	0.364	0.033	0.695	0.662	TRUE
MCAR 30% (achieved 30%)	Complete−case	Shrinkage	2 vs 0	0.364	0.013	0.715	0.702	TRUE
MCAR 30% (achieved 30%)	Complete−case	Non−shrinkage	3 vs 0	0.273	−0.068	0.614	0.682	FALSE
MCAR 30% (achieved 30%)	Complete−case	Shrinkage	3 vs 0	0.273	−0.084	0.630	0.715	FALSE
MCAR 30% (achieved 30%)	Complete−case	Non−shrinkage	4 vs 0	0.393	0.029	0.756	0.727	TRUE
MCAR 30% (achieved 30%)	Complete−case	Shrinkage	4 vs 0	0.393	0.032	0.754	0.721	TRUE
MCAR 30% (achieved 30%)	PMM MI	Non−shrinkage	1 vs 0	0.443	0.083	0.804	0.721	TRUE
MCAR 30% (achieved 30%)	PMM MI	Shrinkage	1 vs 0	0.443	0.090	0.797	0.707	TRUE
MCAR 30% (achieved 30%)	PMM MI	Non−shrinkage	2 vs 0	0.381	0.068	0.694	0.626	TRUE
MCAR 30% (achieved 30%)	PMM MI	Shrinkage	2 vs 0	0.381	0.057	0.705	0.648	TRUE
MCAR 30% (achieved 30%)	PMM MI	Non−shrinkage	3 vs 0	0.275	−0.039	0.590	0.629	FALSE
MCAR 30% (achieved 30%)	PMM MI	Shrinkage	3 vs 0	0.275	−0.046	0.597	0.643	FALSE
MCAR 30% (achieved 30%)	PMM MI	Non−shrinkage	4 vs 0	0.419	0.070	0.768	0.698	TRUE
MCAR 30% (achieved 30%)	PMM MI	Shrinkage	4 vs 0	0.419	0.072	0.766	0.694	TRUE
MCAR 30% (achieved 30%)	Propensity−score MI	Non−shrinkage	1 vs 0	0.419	0.032	0.806	0.773	TRUE
MCAR 30% (achieved 30%)	Propensity−score MI	Shrinkage	1 vs 0	0.419	0.039	0.799	0.761	TRUE
MCAR 30% (achieved 30%)	Propensity−score MI	Non−shrinkage	2 vs 0	0.386	0.047	0.726	0.679	TRUE
MCAR 30% (achieved 30%)	Propensity−score MI	Shrinkage	2 vs 0	0.386	0.038	0.735	0.697	TRUE

IBScovars_primary_30pct_table.csv

scenario	method	variance_method	contrast	estimate	lower	upper	width	significant
MCAR 30% (achieved 30%)	Propensity−score MI	Non−shrinkage	3 vs 0	0.261	−0.072	0.593	0.666	FALSE
MCAR 30% (achieved 30%)	Propensity−score MI	Shrinkage	3 vs 0	0.261	−0.079	0.600	0.680	FALSE
MCAR 30% (achieved 30%)	Propensity−score MI	Non−shrinkage	4 vs 0	0.403	0.009	0.797	0.788	TRUE
MCAR 30% (achieved 30%)	Propensity−score MI	Shrinkage	4 vs 0	0.403	0.008	0.797	0.789	TRUE
MCAR 30% (achieved 30%)	Regression MI	Non−shrinkage	1 vs 0	0.459	0.085	0.833	0.748	TRUE
MCAR 30% (achieved 30%)	Regression MI	Shrinkage	1 vs 0	0.459	0.093	0.825	0.733	TRUE
MCAR 30% (achieved 30%)	Regression MI	Non−shrinkage	2 vs 0	0.402	0.045	0.759	0.714	TRUE
MCAR 30% (achieved 30%)	Regression MI	Shrinkage	2 vs 0	0.402	0.037	0.767	0.730	TRUE
MCAR 30% (achieved 30%)	Regression MI	Non−shrinkage	3 vs 0	0.292	−0.052	0.637	0.689	FALSE
MCAR 30% (achieved 30%)	Regression MI	Shrinkage	3 vs 0	0.292	−0.057	0.642	0.700	FALSE
MCAR 30% (achieved 30%)	Regression MI	Non−shrinkage	4 vs 0	0.436	0.034	0.839	0.804	TRUE
MCAR 30% (achieved 30%)	Regression MI	Shrinkage	4 vs 0	0.436	0.037	0.836	0.800	TRUE

IBScovars_primary_30pct_table.csv

benchmark	error	benchmark_covered	shrink_alpha
0.285	0.122	TRUE	NA
0.285	0.122	TRUE	0.781
0.297	0.067	TRUE	NA
0.297	0.067	TRUE	0.781
0.351	−0.078	TRUE	NA
0.351	−0.078	TRUE	0.781
0.348	0.045	TRUE	NA
0.348	0.045	TRUE	0.781
0.285	0.159	TRUE	NA
0.285	0.159	TRUE	0.521
0.297	0.084	TRUE	NA
0.297	0.084	TRUE	0.521
0.351	−0.075	TRUE	NA
0.351	−0.075	TRUE	0.521
0.348	0.071	TRUE	NA
0.348	0.071	TRUE	0.521
0.285	0.134	TRUE	NA
0.285	0.134	TRUE	0.484
0.297	0.090	TRUE	NA
0.297	0.090	TRUE	0.484

IBScovars_primary_30pct_table.csv

benchmark	error	benchmark_covered	shrink_alpha
0.351	−0.090	TRUE	NA
0.351	−0.090	TRUE	0.484
0.348	0.055	TRUE	NA
0.348	0.055	TRUE	0.484
0.285	0.174	TRUE	NA
0.285	0.174	TRUE	0.523
0.297	0.105	TRUE	NA
0.297	0.105	TRUE	0.523
0.351	−0.058	TRUE	NA
0.351	−0.058	TRUE	0.523
0.348	0.088	TRUE	NA
0.348	0.088	TRUE	0.523

IBScovars_shrinkage_width_gain.csv

missing_rate	method	contrast	Non−shrinkage	Shrinkage	width_reduction_pct
0.1	PMM MI	1 vs 0	0.649	0.644	0.817
0.1	PMM MI	2 vs 0	0.575	0.616	−7.227
0.1	PMM MI	3 vs 0	0.627	0.641	−2.231
0.1	PMM MI	4 vs 0	0.629	0.641	−1.772
0.1	Propensity−score MI	1 vs 0	0.657	0.651	0.960
0.1	Propensity−score MI	2 vs 0	0.585	0.627	−7.271
0.1	Propensity−score MI	3 vs 0	0.633	0.647	−2.128
0.1	Propensity−score MI	4 vs 0	0.638	0.649	−1.627
0.1	Regression MI	1 vs 0	0.670	0.666	0.657
0.1	Regression MI	2 vs 0	0.595	0.637	−7.095
0.1	Regression MI	3 vs 0	0.657	0.669	−1.844
0.1	Regression MI	4 vs 0	0.667	0.678	−1.561
0.2	PMM MI	1 vs 0	0.690	0.692	−0.315
0.2	PMM MI	2 vs 0	0.622	0.665	−6.926
0.2	PMM MI	3 vs 0	0.695	0.704	−1.378
0.2	PMM MI	4 vs 0	0.678	0.688	−1.451
0.2	Propensity−score MI	1 vs 0	0.687	0.686	0.157
0.2	Propensity−score MI	2 vs 0	0.664	0.702	−5.632
0.2	Propensity−score MI	3 vs 0	0.712	0.719	−0.930
0.2	Propensity−score MI	4 vs 0	0.687	0.696	−1.349

IBScovars_shrinkage_width_gain.csv

missing_rate	method	contrast	Non−shrinkage	Shrinkage	width_reduction_pct
0.2	Regression MI	1 vs 0	0.715	0.716	−0.070
0.2	Regression MI	2 vs 0	0.657	0.697	−6.147
0.2	Regression MI	3 vs 0	0.705	0.713	−1.204
0.2	Regression MI	4 vs 0	0.733	0.738	−0.786
0.3	PMM MI	1 vs 0	0.721	0.707	1.973
0.3	PMM MI	2 vs 0	0.626	0.648	−3.523
0.3	PMM MI	3 vs 0	0.629	0.643	−2.258
0.3	PMM MI	4 vs 0	0.698	0.694	0.531
0.3	Propensity−score MI	1 vs 0	0.773	0.761	1.655
0.3	Propensity−score MI	2 vs 0	0.679	0.697	−2.700
0.3	Propensity−score MI	3 vs 0	0.666	0.680	−2.078
0.3	Propensity−score MI	4 vs 0	0.788	0.789	−0.058
0.3	Regression MI	1 vs 0	0.748	0.733	2.117
0.3	Regression MI	2 vs 0	0.714	0.730	−2.331
0.3	Regression MI	3 vs 0	0.689	0.700	−1.585
0.3	Regression MI	4 vs 0	0.804	0.800	0.603
0.4	PMM MI	1 vs 0	0.699	0.688	1.597
0.4	PMM MI	2 vs 0	0.718	0.729	−1.615
0.4	PMM MI	3 vs 0	0.698	0.713	−2.195
0.4	PMM MI	4 vs 0	0.690	0.708	−2.713

IBScovars_shrinkage_width_gain.csv

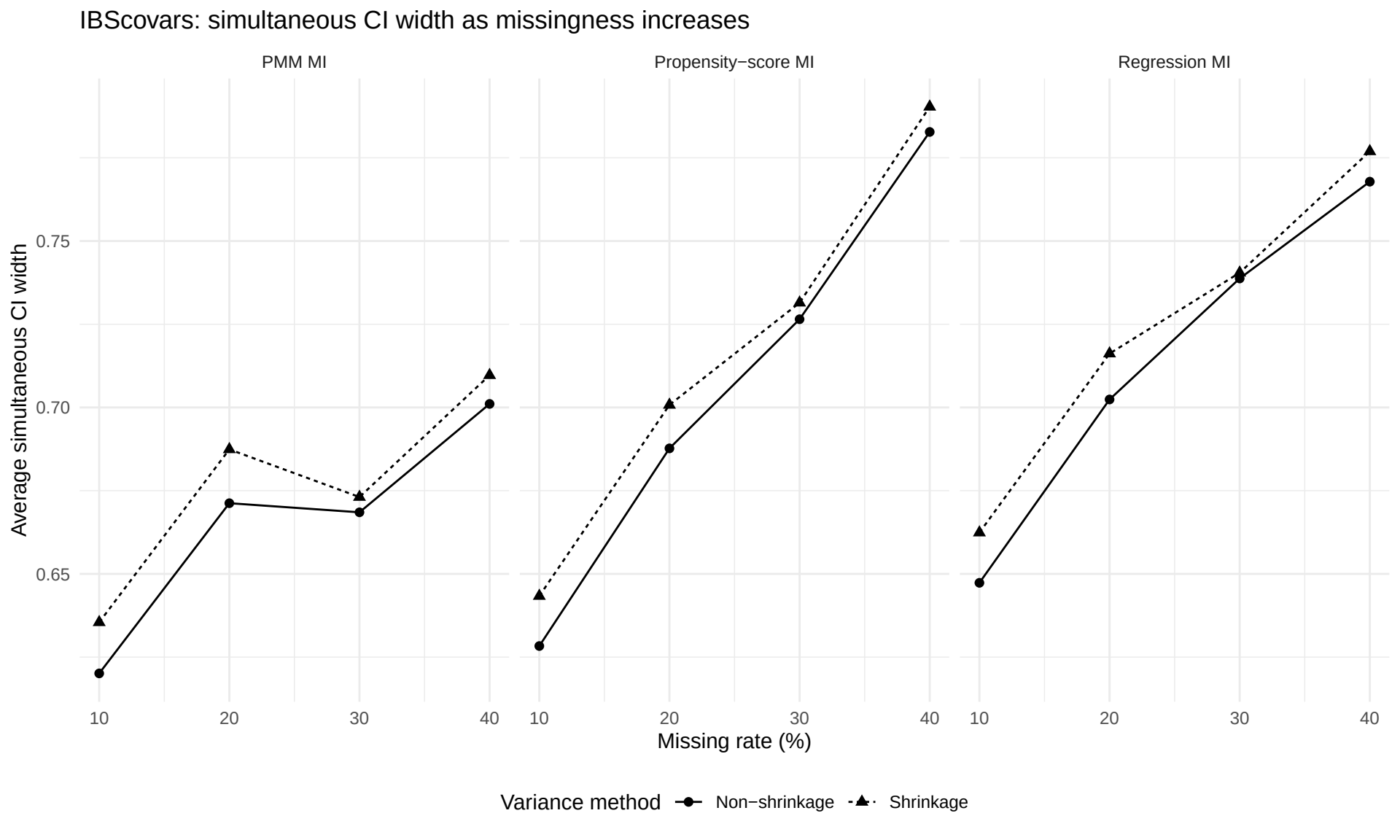
missing_rate	method	contrast	Non−shrinkage	Shrinkage	width_reduction_pct
0.4	Propensity−score MI	1 vs 0	0.830	0.823	0.762
0.4	Propensity−score MI	2 vs 0	0.786	0.795	−1.126
0.4	Propensity−score MI	3 vs 0	0.780	0.790	−1.162
0.4	Propensity−score MI	4 vs 0	0.735	0.754	−2.533
0.4	Regression MI	1 vs 0	0.814	0.809	0.603
0.4	Regression MI	2 vs 0	0.783	0.794	−1.416
0.4	Regression MI	3 vs 0	0.718	0.731	−1.854
0.4	Regression MI	4 vs 0	0.757	0.774	−2.235

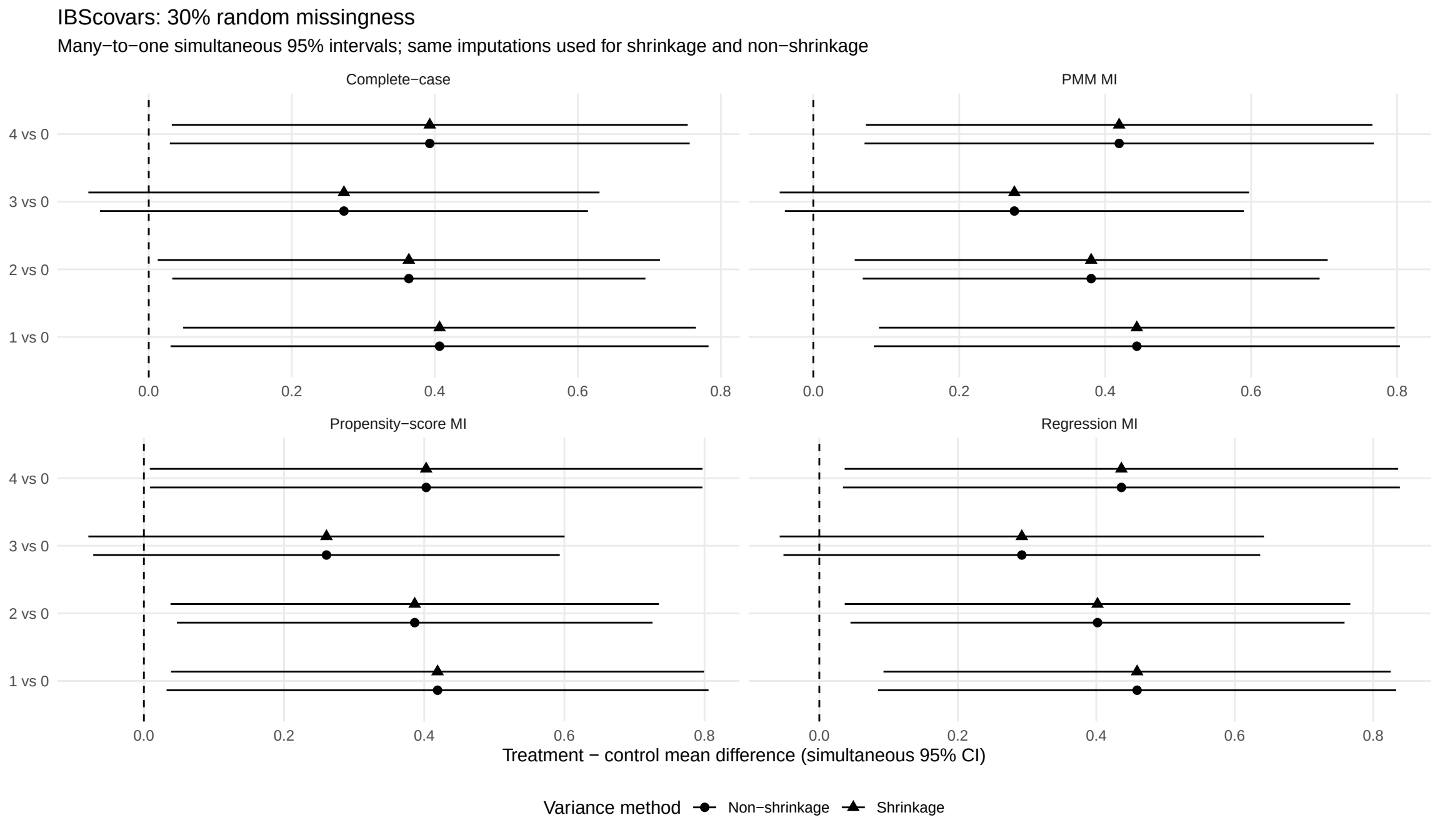
IBScovars_validation_summary.csv

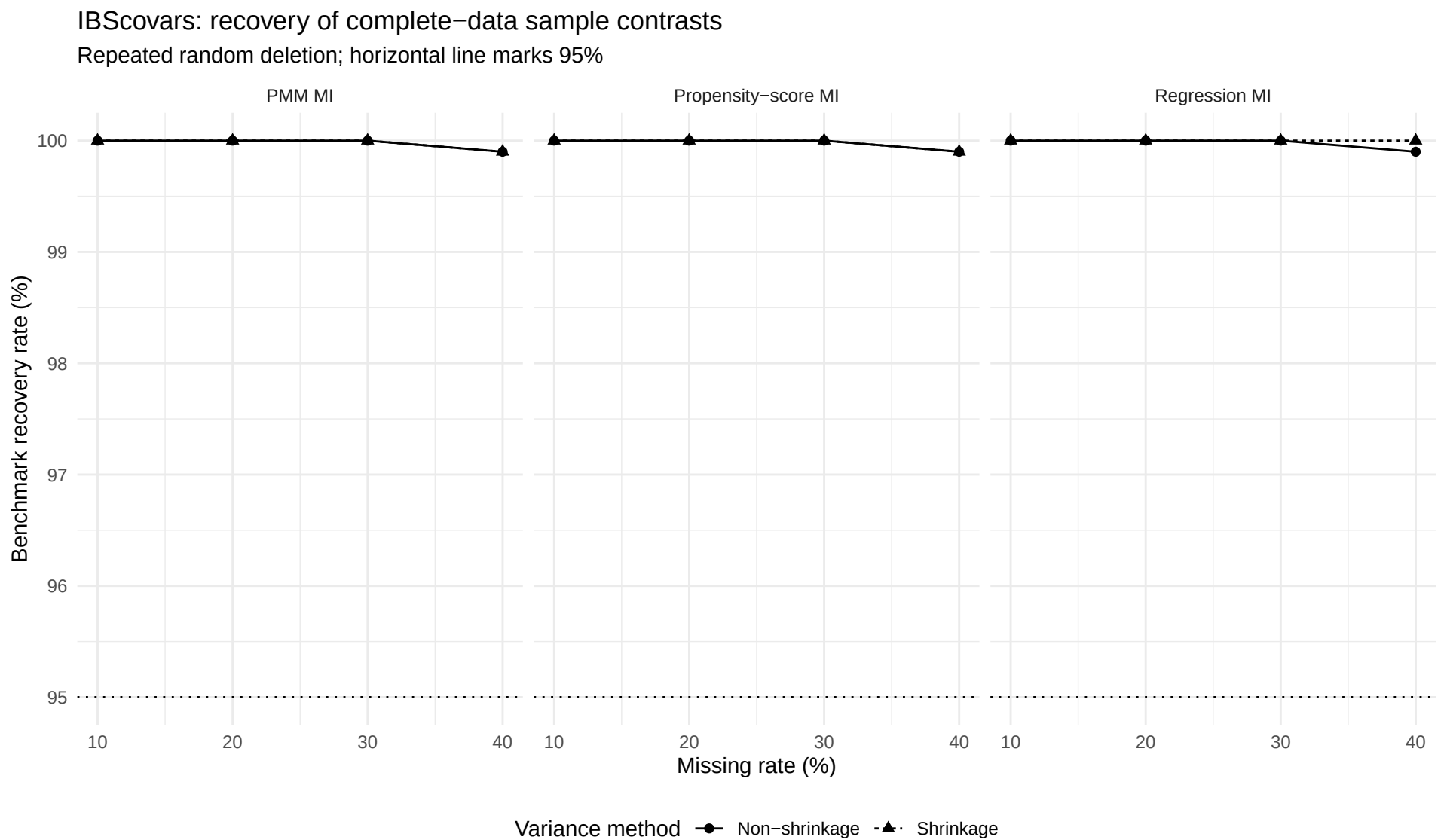
missing_rate	method	variance_method	n_rows	rmse	mean_abs_error	benchmark_recovery_pct	avg_ci_width	mean_shrink_alpha
0.1	PMM MI	Non−shrinkage	1000	0.040	0.031	100.0	0.627	NA
0.1	PMM MI	Shrinkage	1000	0.040	0.031	100.0	0.641	0.828
0.1	Propensity−score MI	Non−shrinkage	1000	0.040	0.032	100.0	0.630	NA
0.1	Propensity−score MI	Shrinkage	1000	0.040	0.032	100.0	0.643	0.825
0.1	Regression MI	Non−shrinkage	1000	0.040	0.032	100.0	0.635	NA
0.1	Regression MI	Shrinkage	1000	0.040	0.032	100.0	0.649	0.826
0.2	PMM MI	Non−shrinkage	1000	0.064	0.051	100.0	0.654	NA
0.2	PMM MI	Shrinkage	1000	0.064	0.051	100.0	0.666	0.708
0.2	Propensity−score MI	Non−shrinkage	1000	0.065	0.051	100.0	0.667	NA
0.2	Propensity−score MI	Shrinkage	1000	0.065	0.051	100.0	0.678	0.705
0.2	Regression MI	Non−shrinkage	1000	0.065	0.051	100.0	0.679	NA
0.2	Regression MI	Shrinkage	1000	0.065	0.051	100.0	0.690	0.699
0.3	PMM MI	Non−shrinkage	1000	0.079	0.063	100.0	0.682	NA
0.3	PMM MI	Shrinkage	1000	0.079	0.063	100.0	0.692	0.627
0.3	Propensity−score MI	Non−shrinkage	1000	0.078	0.062	100.0	0.710	NA
0.3	Propensity−score MI	Shrinkage	1000	0.078	0.062	100.0	0.719	0.619
0.3	Regression MI	Non−shrinkage	1000	0.078	0.062	100.0	0.732	NA
0.3	Regression MI	Shrinkage	1000	0.078	0.062	100.0	0.740	0.610
0.4	PMM MI	Non−shrinkage	1000	0.103	0.082	99.9	0.711	NA
0.4	PMM MI	Shrinkage	1000	0.103	0.082	99.9	0.719	0.556

IBScovars_validation_summary.csv

missing_rate	method	variance_method	n_rows	rmse	mean_abs_error	benchmark_recovery_pct	avg_ci_width	mean_shrink_alpha
0.4	Propensity−score MI	Non−shrinkage	1000	0.103	0.082	99.9	0.763	NA
0.4	Propensity−score MI	Shrinkage	1000	0.103	0.082	99.9	0.770	0.548
0.4	Regression MI	Non−shrinkage	1000	0.104	0.082	99.9	0.798	NA
0.4	Regression MI	Shrinkage	1000	0.104	0.082	100.0	0.805	0.533


IBScovars: simultaneous CI width as missingness increases
PMM MI
Propensity−score MI
Regression MI
Average simultaneous CI width
0.75
0.70
0.65
10
20
30
40
Missing rate (%)
Variance method
Non−shrinkage
Shrinkage


IBScovars: 30% random missingness
Many−to−one simultaneous 95% intervals; same imputations used for shrinkage and non−shrinkage
Complete−case
PMM MI
Propensity−score MI
Regression MI
4 vs 0
3 vs 0
2 vs 0
1 vs 0
0.0
0.2
0.4
0.6
0.8
Treatment − control mean difference (simultaneous 95% CI)
Variance method
Non−shrinkage
Shrinkage


IBScovars: recovery of complete−data sample contrasts
Repeated random deletion; horizontal line marks 95%
PMM MI
Propensity−score MI
Regression MI
Benchmark recovery rate (%)
100
99
98
97
96
95
10
20
30
40
Missing rate (%)
Variance method
Non−shrinkage
Shrinkage